

Supplementary Information

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1 Explicit transition matrices for the two-player game

In each subsection below, states are ordered $\mathbf{a} = (0, 0)$, $\mathbf{b} = (0, 1)$, $\mathbf{c} = (1, 0)$, $\mathbf{d} = (1, 1)$; rows and columns of each matrix correspond to these states in the same order.

Moran process

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Now, by taking the fundamental matrix of T and performing a right multiplication by R , the matrix with entries which represent transitions from transitive states to absorbing states[?], we can obtain the absorption matrix for this system.

For the above game, we find the following equation:

$$\left(I - \begin{bmatrix} 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 \\ 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix} \quad (2)$$

And solve to find our absorption matrix:

$$\begin{bmatrix} \frac{\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}}{\left(\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}\right)(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} & \frac{\frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}}{\left(\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}\right)(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} \\ \frac{\frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}}{\left(\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}\right)(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} & \frac{\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}}{\left(\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}\right)(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} \end{bmatrix} \quad (3)$$

In this matrix, the rows represent our initial transitive state (in our system, row 1 represents $\mathbf{b}$ and row 2 represents $\mathbf{c}$), and our columns represent which state the system is absorbed into (with $\mathbf{a}$ on the left, and $\mathbf{d}$ on the right). It is immediately clear from this result that the probability of the system ending up in a particular absorbing state does not rely on the fitness of individuals in the absorbing state, but rather their fitness in the transitive states. Thus, players can remain in a state which could be bettered by a single change, however they will not make such a change if no individual of another type exists in the population.

Fermi imitation dynamics

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2}\phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \frac{1}{2} & 0 & \frac{1}{2}\phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \frac{1}{2}\phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & 0 & \frac{1}{2} & \frac{1}{2}\phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

For the two-player state space $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$, the explicit transition matrix is given in equation (4) in the supplementary material. Applying the same absorption matrix calculation as for the Moran process, we obtain:

$$\begin{bmatrix} \phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & \phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \end{bmatrix} \quad (5)$$

This shows that, similarly to the Moran process, our absorption probabilities rely on the fitness of players in transitive states and not those of players in the absorbing states. Fermi imitation dynamics is therefore poorly suited to heterogeneous populations: a player may copy a high-fitness strategy without considering whether that strategy would benefit them, and may even enter an absorbing state that worsens their own payoff.

Aspiration Dynamics

The transition matrix for the two-player, two-action case with state space $\mathbf{a} = (0, 0)$, $\mathbf{b} = (0, 1)$, $\mathbf{c} = (1, 0)$, $\mathbf{d} = (1, 1)$ is:

$$\begin{bmatrix} 1 - \frac{1}{2(1+e^{-\beta(A_2-J_2(a))})} - \frac{1}{2(1+e^{-\beta(A_1-J_1(a))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta J_2(a)})} & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta J_1(a)})} & 0 \\ \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta J_2(b)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2-J_2(b))})} - \frac{1}{2(1+e^{-\beta(A_1-J_1(b))})} & 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta J_1(b)})} \\ \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta J_1(c)})} & 0 & 1 - \frac{1}{2(1+e^{-\beta(A_2-J_2(c))})} - \frac{1}{2(1+e^{-\beta(A_1-J_1(c))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta J_2(c)})} \\ 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta J_1(d)})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta J_2(d)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2-J_2(d))})} - \frac{1}{2(1+e^{-\beta(A_1-J_1(d))})} \end{bmatrix} \quad (6)$$

We can see that as our players no longer consider the actions of the rest of the population when considering their decision, we no longer have an absorbing Markov chain. Thus, we no longer consider an absorption matrix, but instead we consider the steady state of our Markov chain [?]. This is the proportion of time spent in each state when the Markov chain reaches a stable distribution across the states. Any irreducible, aperiodic, finite Markov chain has a unique stationary distribution, and the chain converges to it from any starting state [?]. For our four state system, we acquire the following result for a stationary distribution:

Introspective Dynamics

The transition matrix for the two-player, two-action case with state space $\mathbf{a} = (0, 0)$, $\mathbf{b} = (0, 1)$, $\mathbf{c} = (1, 0)$, $\mathbf{d} = (1, 1)$, following Couto [?], is:

$$\begin{bmatrix} 1 - \frac{1}{2(e^{\beta(\pi_2(a)-\pi_2(b))+1})} - \frac{1}{2(e^{\beta(\pi_1(a)-\pi_1(c))+1})} & \frac{1}{2(e^{\beta(\pi_2(a)-\pi_2(b))+1})} & \frac{1}{2(e^{\beta(\pi_1(a)-\pi_1(c))+1})} & 0 \\ \frac{1}{2(e^{\beta(-\pi_2(a)+\pi_2(b))+1})} & 1 - \frac{1}{2(e^{\beta(\pi_1(b)-\pi_1(d))+1})} - \frac{1}{2(1+e^{-\beta(\pi_2(a)-\pi_2(b))})} & 0 & \frac{1}{2(e^{\beta(\pi_1(b)-\pi_1(d))+1})} \\ \frac{1}{2(e^{\beta(-\pi_1(a)+\pi_1(c))+1})} & 0 & 1 - \frac{1}{2(e^{\beta(\pi_2(c)-\pi_2(d))+1})} - \frac{1}{2(1+e^{-\beta(\pi_1(a)-\pi_1(c))})} & \frac{1}{2(e^{\beta(\pi_2(c)-\pi_2(d))+1})} \\ 0 & \frac{1}{2(e^{\beta(-\pi_1(b)+\pi_1(d))+1})} & \frac{1}{2(e^{\beta(-\pi_2(c)+\pi_2(d))+1})} & 1 - \frac{1}{2(1+e^{-\beta(\pi_2(c)-\pi_2(d))})} - \frac{1}{2(1+e^{-\beta(\pi_1(b)-\pi_1(d))})} \end{bmatrix} \quad (7)$$

We then consider the steady state vectors for this system. For this, we take the left eigenvector of the transition matrix, which is given by:

Unlike in the case of imitation dynamics, there are no absorbing states for a Markov chain transitioning according to imitation dynamics. Therefore, we define a steady state vector $\mathbf{u}$ according to the method Couto uses in [?].

Introspective Imitation Dynamics

Note that Introspective Imitation Dynamics takes into account both selection intensity and choice intensity. This means that both the level to which a player's fitness is considered when their strategy is being chosen for consideration, and also how much it is considered when choosing whether to accept the new strategy, is present in the dynamic.

Let $\eta_{i,j}(\mathbf{x}, \mathbf{y}) = 1 + e^{\beta(f_i(\mathbf{x}) - f_j(\mathbf{y}))}$ be the denominator of the Fermi function. For the two-player state space $S = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}\}$, the explicit transition matrix is given in equation (9) in the supplementary material. Applying the absorption matrix calculation, we obtain:

$$\begin{bmatrix} \frac{\eta_{1,1}(\mathbf{b}, \mathbf{d})f_1(b)}{\eta_{1,1}(\mathbf{b}, \mathbf{d})f_1(b) + \eta_{2,2}(\mathbf{b}, \mathbf{a})f_2(b)} & \frac{\eta_{2,2}(\mathbf{b}, \mathbf{a})f_2(b)}{\eta_{1,1}(\mathbf{b}, \mathbf{d})f_1(b) + \eta_{2,2}(\mathbf{b}, \mathbf{a})f_2(b)} \\ \frac{\eta_{2,2}(\mathbf{c}, \mathbf{d})f_2(c)}{\eta_{1,1}(\mathbf{c}, \mathbf{a})f_1(c) + \eta_{2,2}(\mathbf{c}, \mathbf{d})f_2(c)} & \frac{\eta_{1,1}(\mathbf{c}, \mathbf{a})f_1(c)}{\eta_{1,1}(\mathbf{c}, \mathbf{a})f_1(c) + \eta_{2,2}(\mathbf{c}, \mathbf{d})f_2(c)} \end{bmatrix} \quad (8)$$

Unlike other imitation dynamics, we can see that the absorption probability in an introspective imitation process relies on the fitness that at least one individual will possess in the absorbing state. This means that players will enter absorbing states with a higher probability if it improves their own fitness, despite any heterogeneity between payoffs for the same

action played by different players. This solves the problem of players making worsening moves based on the fitness of another individual in the population, while still retaining the property of imitation dynamics that new strategies cannot be introduced to the system.

Let $\eta_{i,j}(\mathbf{x}, \mathbf{y}) = 1 + e^{\beta(f_i(\mathbf{x}) - f_j(\mathbf{y}))}$.

$$\begin{bmatrix} \frac{1}{\frac{f_1(b)}{2\eta_{2,2}(\mathbf{b}, \mathbf{a})(f_1(b)+f_2(b))}} & 0 & 0 & 0 \\ \frac{f_2(c)}{2\eta_{1,1}(\mathbf{c}, \mathbf{a})(f_1(c)+f_2(c))} & 1 - \frac{f_2(b)}{\eta_{1,1}(\mathbf{b}, \mathbf{d})(2f_1(b)+2f_2(b))} - \frac{f_1(b)}{\eta_{2,2}(\mathbf{b}, \mathbf{a})(2f_1(b)+2f_2(b))} & 0 & \frac{f_2(b)}{2\eta_{1,1}(\mathbf{b}, \mathbf{d})(f_1(b)+f_2(b))} \\ 0 & 0 & 0 & \frac{f_1(c)}{2\eta_{2,2}(\mathbf{c}, \mathbf{d})(f_1(c)+f_2(c))} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9)$$

Moran process with mutation

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} + 1 & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} & -\frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} + 1 \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix} \quad (10)$$

Fermi imitation dynamics with mutation

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & -\frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix} \quad (11)$$

Aspiration dynamics with mutation

$$\begin{bmatrix} 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(a))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(a))}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(a)}} & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(a)}} & 0 \\ \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(b)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(b))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(b))}} & 0 & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(b)}} \\ \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(c)}} & 0 & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(c)}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(d)}} \\ 0 & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(d)}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(d)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(d))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(d))}} \end{bmatrix} \quad (12)$$

Introspection dynamics with mutation

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & \frac{\mu_{12}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & -\frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(b)-f_2(b))}+1)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} \\ 0 & \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} & \frac{\mu_{21}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(e^{\beta(f_1(c)-f_2(c))}+1)} \end{bmatrix} \quad (13)$$

Introspective Imitation Dynamics with mutation

Let $\eta_{i,j}(\mathbf{x}, \mathbf{y}) = 1 + e^{\beta(f_i(\mathbf{x}) - f_j(\mathbf{y}))}$.

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{b}, \mathbf{a})(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{b}, \mathbf{a})(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2\eta_{1,1}(\mathbf{b}, \mathbf{d})(\epsilon f_1(b)+\epsilon f_2(b)+2)} + 1 & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2\eta_{1,1}(\mathbf{b}, \mathbf{d})(\epsilon f_1(b)+\epsilon f_2(b)+2)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2\eta_{1,1}(\mathbf{c}, \mathbf{a})(\epsilon f_1(c)+\epsilon f_2(c)+2)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{c}, \mathbf{d})(\epsilon f_1(c)+\epsilon f_2(c)+2)} & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{c}, \mathbf{d})(\epsilon f_1(c)+\epsilon f_2(c)+2)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix} \quad (14)$$

2 Mutation

2.1 The Moran Process

In the case of the Moran process, our transition matrix becomes:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{\sum_{a_i=b_{I(\mathbf{a},\mathbf{b})}} f(a_i)}{N \sum_{a_j} f(a_j)} \right) (1 - \sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + \left(\frac{\mu_{i,\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \text{ differing at position } I(\mathbf{a}, \mathbf{b}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (15)$$

The explicit transition matrix for the two-player case is given in equation (10) in the supplementary material. This yields an ergodic Markov chain whose steady state is:

2.2 Fermi Imitation Dynamics

When we introduce mutation to Fermi Imitation Dynamics, we obtain the following transition matrix:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N(N-1)} \sum_{a_j=b_{I(\mathbf{a},\mathbf{b})}} \phi(\pi_{I(\mathbf{a}, \mathbf{b})}(a) - \pi_j(\mathbf{a})) \right) (1 - \sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + \left(\frac{\mu_{i,\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (16)$$

The explicit transition matrix for the two-player case is given in equation (11) in the supplementary material. The resulting steady state is:

2.3 Aspiration Dynamics

As aspiration and introspection dynamics do not form absorbing Markov chains, we do not need mutation to be applied in order to form an ergodic chain. However, we will consider these cases for completeness, and thus we obtain the following transition matrix:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N} \phi(\pi_{I(\mathbf{a},\mathbf{b})} - A_i) \right) (1 - \sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + \left(\frac{\mu_{i,\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a}, \mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases}$$

The explicit transition matrix for the two-player case is given in equation (12) in the supplementary material.

2.4 Introspection Dynamics

As introspection dynamics does not form an absorbing Markov chain, we do not need mutation to form an ergodic chain. However, we will consider the following transition matrix for the case with mutation:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N(m_j - 1)} \phi(\Delta \pi_{I(\mathbf{a},\mathbf{b})}) \right) (1 - \sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + \left(\frac{\mu_{i,\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a}, \mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases}$$

The explicit transition matrix for the two-player case is given in equation (13) in the supplementary material.

2.5 Introspective Imitation Dynamics

Finally, for the case of introspective imitation dynamics, we get the following transition matrix with mutation:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N} \frac{\sum_{a_j=b} f_j(\mathbf{a}, \mathbf{b})}{\sum_k f_k(\mathbf{a})} \phi(\Delta(f_{I(\mathbf{a}, \mathbf{b})})) (1 - \sum_k \mu_{I(\mathbf{a}, \mathbf{b}), k}) + \left(\frac{\mu_{i, \mathbf{b}_{I(\mathbf{a}, \mathbf{b})}}}{N} \right) \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (17)$$

The explicit transition matrix for the two-player case is given in equation (14) in the supplementary material. The resulting steady state is:

2.6 Extended parameter sweeps

Fermi imitation dynamics

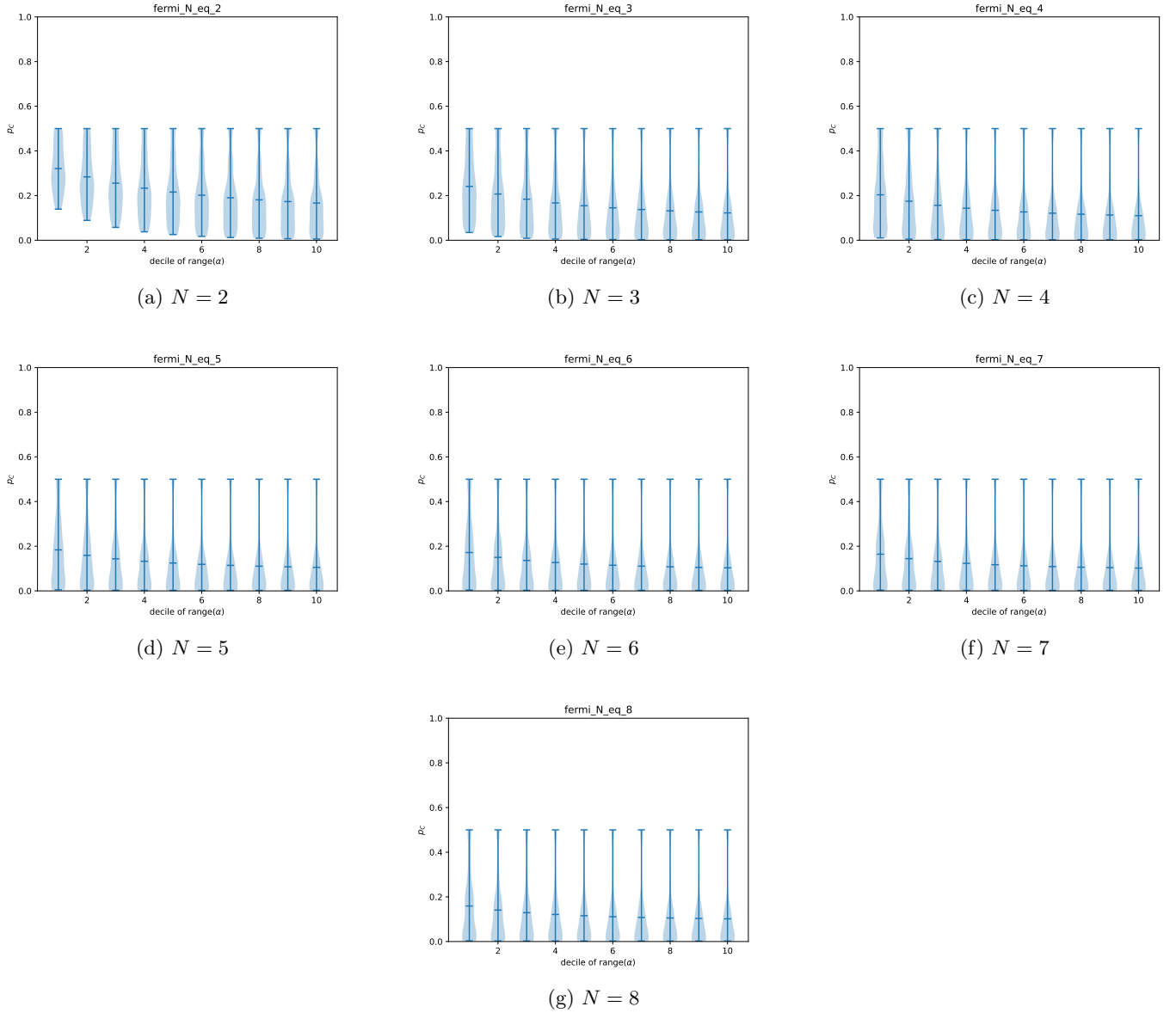
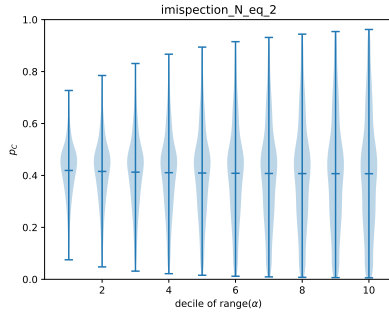
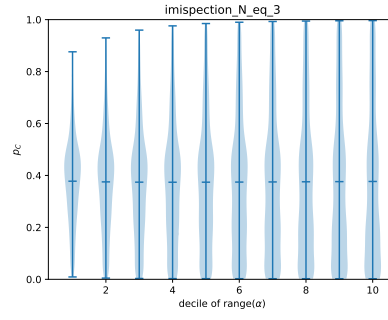


Figure 1: Fermi imitation dynamics across population sizes N .

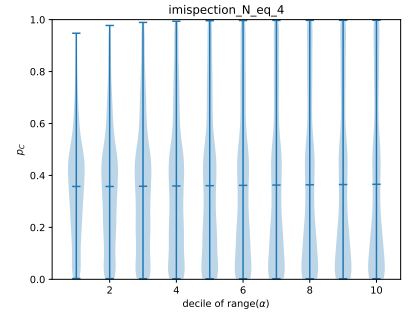
Introspective Imitation Dynamics



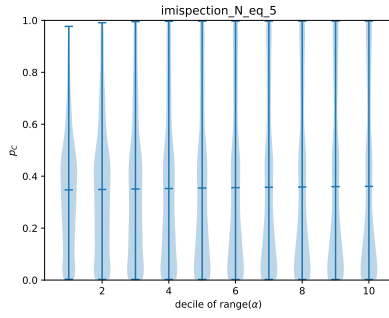
(a) $N = 2$



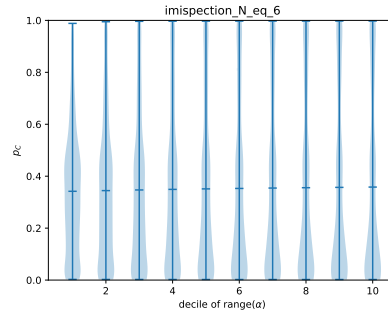
(b) $N = 3$



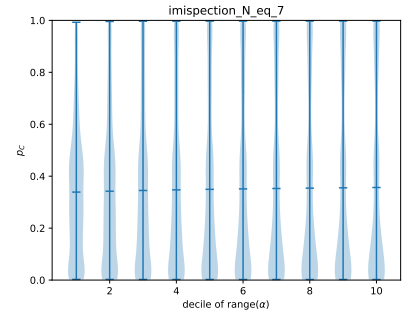
(c) $N = 4$



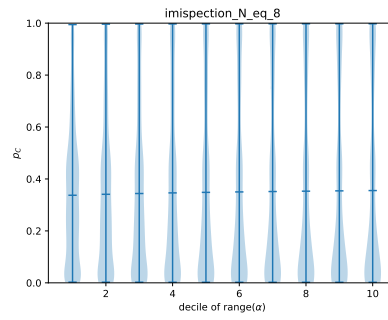
(d) $N = 5$



(e) $N = 6$



(f) $N = 7$

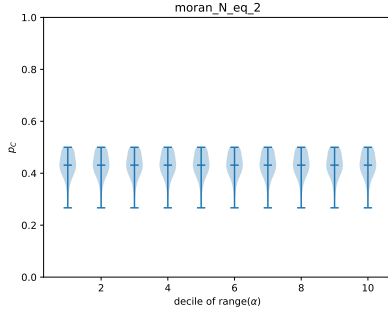


(g) $N = 8$

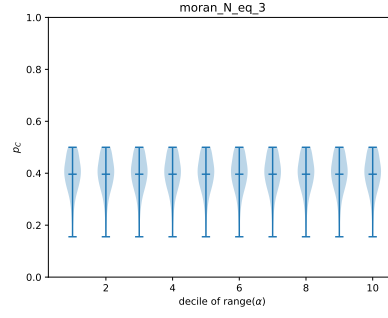
Figure 2: Introspective imitation dynamics across population sizes N .

Moran process

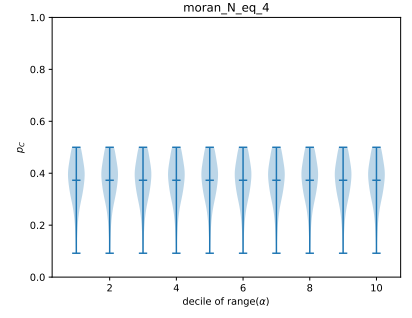
Introspection dynamics



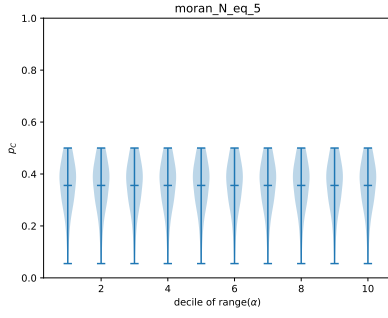
(a) $N = 2$



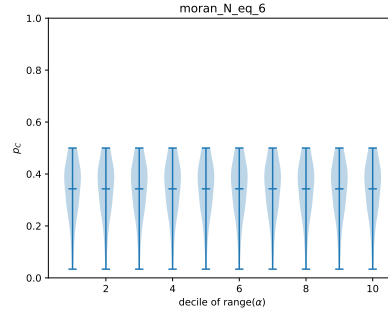
(b) $N = 3$



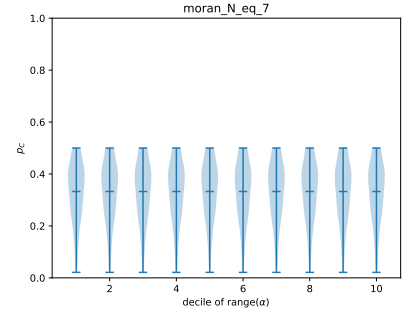
(c) $N = 4$



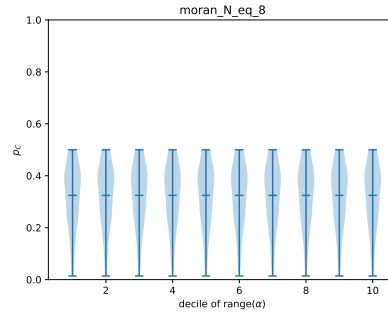
(d) $N = 5$



(e) $N = 6$

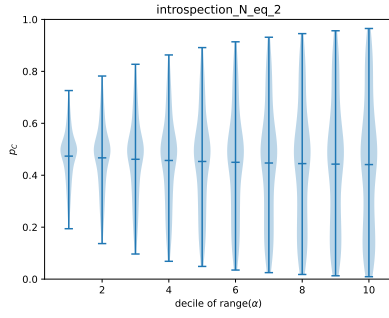


(f) $N = 7$

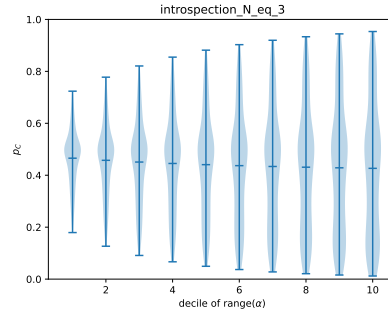


(g) $N = 8$

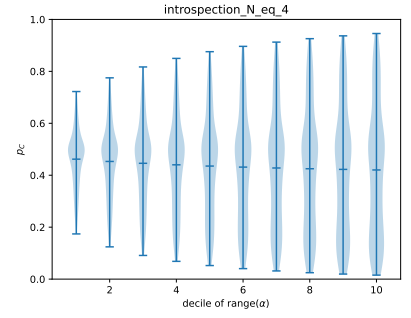
Figure 3: Moran process across population sizes N .



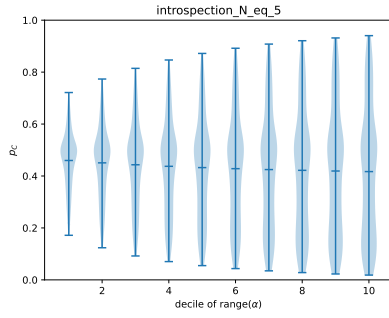
(a) $N = 2$



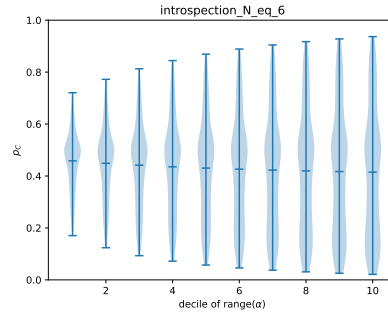
(b) $N = 3$



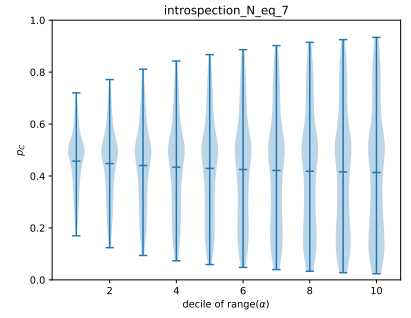
(c) $N = 4$



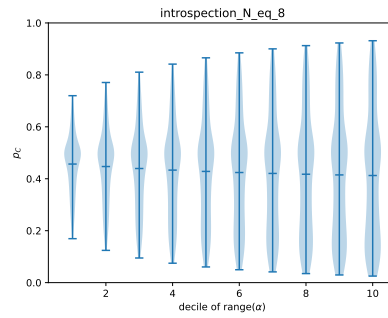
(d) $N = 5$



(e) $N = 6$



(f) $N = 7$



(g) $N = 8$

Figure 4: Introspection dynamics across population sizes N .